6.1.1 Understanding behavior tree execution

Understanding how behavior trees execute is crucial to designing and implementing behavior trees. Unlike most concepts in computer science, behavior trees operate in terms of success and failure. When a node in a behavior tree executes, it will return either success or failure; this even applies to conditions and selector nodes.

Behavior trees execute from top to bottom and left to right. Figure 6.2 shows the process and what happens if a node fails or succeeds. In the example, the AI the tree controls has an apple but no pear. In the first sequence node, a condition checks if the AI has an apple. Because the AI doesn’t have an apple, it aborts the sequence and falls back to the selector. The selector then selects its next child node, another sequence, that checks if the AI has a pear, and because it does, the AI eats the apple.